

Linux Kernel Assignment - Answers

1. What is the Linux Kernel?

The Linux kernel is the core part of the Linux operating system. It acts as a bridge between hardware and software. It manages CPU, memory, storage, processes, and communication with hardware devices. Without the kernel, the operating system cannot function.

2. What is a Kernel Module?

A kernel module is a piece of code that can be loaded into or removed from the Linux kernel while the system is running. It is commonly used for device drivers and helps extend kernel functionality without rebooting the system.

3. Why Should Beginners Use a VM for Kernel Experiments?

A Virtual Machine provides a safe environment for learning kernel development. If a mistake crashes the operating system, only the VM is affected. Snapshots can be used to restore the system quickly.

4. What is the Purpose of the /proc Folder?

The /proc directory is a virtual filesystem that provides information about the kernel and running processes. It contains details about CPU usage, memory usage, processes, and other system information.

5. Open the /dev Folder and Try to Guess: Which Files Are Related to Your Hard Drive? Which Might Belong to Your Keyboard or Mouse?

Hard-drive related files are usually /dev/sda, /dev/sda1, /dev/sda2, /dev/nvme0n1, etc. Keyboard and mouse devices are commonly found under /dev/input, such as /dev/input/event0, /dev/input/event1, /dev/input/mouse0, and /dev/input/mice.